Confidence-Gated Dual-Process Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable Inertial Sensor Data

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***Abstract*—Wearable fall detectors must trade missed falls against false alarms, and their errors concentrate on everyday movements that resemble falls. This study evaluates a confidence-gated decision layer around a fixed detector under a sensor-only protocol. A frozen convolutional, recurrent and attention-based detector scores every sensor window. A two-threshold confidence gate, calibrated under a cost that weights a missed fall ten times a false alarm, decides confident windows directly and escalates ambiguous ones to a deliberative path that summarises biomechanical evidence, retrieves similar training cases and queries a locally hosted language model. In five subject-independent SisFall folds, the best blend achieved F1 0.772 ± 0.025 versus 0.768 ± 0.019 for the frozen detector, with expected cost 1079.6 ± 115.3 versus 1086.6 ± 80.6. Event-level and prospective free-living validation remain necessary.**

***Index Terms*—Clinical decision support, confidence gating, cost-sensitive learning, fall detection, large language models, retrieval-augmented reasoning, selective prediction, wearable inertial sensors.**

I. Introduction

FALLS are the second leading cause of unintentional-injury death worldwide, with an estimated 684,000 fatalities and 37.3 million falls requiring medical attention each year; death rates are highest above age 60 [1]. Time spent on the floor unable to summon help—the *long lie*—is independently associated with hospitalisation and institutionalisation [2], [3]. Wearable inertial measurement units (IMUs) are therefore attractive for continuous home monitoring [4]–[7], and deep detectors now report strong benchmark accuracy [8], [9].

Three gaps remain. *First*, stumbles and compensatory recoveries can resemble falls in wearable signals [10]–[12]. We therefore evaluate a stress set containing stumbles, non-fall movements, falls, and other ADLs. *Second*, softmax scores are a poor basis for deferral: modern networks are miscalibrated [13], [14], and the decision to abstain needs its own risk analysis [15]–[18]. *Third*, a fall/ADL bit is not a response: dispatch, caregiver notification, monitoring and logging carry unequal costs, and missed falls dominate risk [19], [20].

Bhatti *et al.* [21] present a hybrid CNN–LSTM–Attention detector for multi-phase fall analysis. That line of work optimises the detector. We ask a different question: *holding the detector fixed, can a decision layer raise recall and lower expected response cost on the hard cases?* Our answer is the Risk-Calibrated Dual-Process (RCDP) pipeline (Fig. 1). System 1 is the frozen primary detector plus a dual-threshold gate. System 2 handles only ambiguous windows: it serialises biomechanical evidence, retrieves labelled training cases [22], [23], queries a local language model [24]–[27] and emits a simulated graded response. Classification is scored under

$$\mathcal{R}_\beta = \beta \cdot \mathrm{FN} + \mathrm{FP}, \qquad \beta = 10. \tag{1}$$

a) Contributions:

- A selective-risk decomposition (Prop. 2) stating when escalation to an adjudicator lowers risk with edge decisions held fixed.
- A dual-process decision layer—gate, evidence, retrieval, local reasoning, simulated action—around a primary detector whose weights are never updated (Fig. 1, Algs. 1–2).
- A subject-independent evaluation on SisFall (main and stress sets) and KFall (binary and stress sets), with a second backbone, a 2×2 retrieval/reasoner ablation, a cost-derived threshold control and constant-prediction controls.
- Two negative results: retrieval or reasoning alone does not reproduce the gain, and a critic permitted to overturn fall labels multiplies missed falls.

All gains are measured against the *same* primary-detector checkpoints under our protocol; we do not compare with the published multi-phase KFall results of [21], which use a different protocol and label granularity.

II. Related Work

Table I positions RCDP against adjacent literature families.

A. Wearable Fall Detection

Threshold rules on acceleration magnitude [5], [28] gave way to classical learners and then to CNN, LSTM, TCN and Transformer models evaluated on SisFall [7], KFall [29], UP-Fall [36], UniMiB SHAR [37] and FallAllD [38]. Surveys identify the free-living trade-off between sensitivity and nuisance alarms as the unsolved problem [4], [6], [8], [9], and Noury *et al.* [20] argue that detectors should be evaluated under a declared cost and protocol. We reuse the CNN–LSTM–Attention motif of [21] as the primary detector and place the novelty at decision time.

TABLE I
CONCEPTUAL POSITIONING AGAINST SELECTED ADJACENT LITERATURE FAMILIES. THE COMPARISON MOTIVATES THE EVALUATED COMBINATION OF DEFERRAL, RETRIEVED EVIDENCE, LANGUAGE-MODEL ADJUDICATION AND SIMULATED RESPONSE; IT IS NOT AN EXHAUSTIVE NOVELTY SEARCH OR EVIDENCE THAT NO PRIOR SYSTEM COMBINES THESE ELEMENTS.

Family	Uncertainty handling	Explanation	Response policy	Near-fall protocol	Asymmetric cost
Threshold rules [5], [28]	fixed magnitude cut	rule trace	binary alarm	no	implicit
Deep IMU detectors [8], [21], [29]	softmax score	attention maps	none	partial (phase labels)	no
Selective prediction / learning to defer [15], [17], [18], [30]	learned reject function	none	none	no	optional
Split conformal prediction [31], [32]	coverage-controlled sets	none	none	no	no
LLMs over raw sensor data [33]–[35]	prompt confidence	natural-language rationale	none	no	no
RCDP (this work)	**cost-calibrated dual threshold**	**biomechanical evidence + retrieved cases + rationale**	**graded action agent**	**Near-fall stress benchmark**	$\mathcal{R}_\beta$, β=10

B. Selective Prediction and Deferral

Classification with a reject option dates to Chow [15] and has been developed into selective classification [16], [17], learning with rejection [18] and learning to defer to an expert [30], [39]. Calibration is a prerequisite for thresholding scores [13], [40]. Our gate is a post-hoc, cost-calibrated member of this family; we do not train a joint predictor–rejector because the detector is fixed by design. Split conformal prediction [31], [32], [41], [42] offers set-valued alternatives, but its marginal guarantee requires exchangeability, which subject shift and overlapping windows violate; we leave it to future work. Cost-sensitive learning follows Elkan [19].

C. Language Models over Sensor Data

Retrieval-augmented generation [23] with compact sentence encoders [22], [43] supports case-based prompting; chain-of-thought [24], reason–act loops [25] and self-refinement [44] structure decisions. HARGPT reads raw IMU streams zero-shot [33], SensorLLM aligns sensor channels with text [34] and Health-LLM predicts health outcomes from wearable data [35]. We differ in two ways: the model receives a serialised evidence string rather than raw signals, which limits but does not remove the risk of unsupported rationales [45], [46]; and it is invoked only on the ambiguous band under an explicit clinical cost.

III. PROBLEM FORMULATION

Let $(\mathbf{x}, y) \sim P$ over $\mathcal{X} \times \{0,1\}$, where $\mathbf{x} \in \mathbb{R}^{C\times T}$ is an inertial window and y=1 denotes a fall. The asymmetric loss is

$$\ell_\beta(\hat{y}, y) = \beta \cdot \mathbf{1}\{\hat{y}{=}0, y{=}1\} + \mathbf{1}\{\hat{y}{=}1, y{=}0\}, \tag{2}$$

the population form of (1). A *selective decision system* (f, g, h) comprises a frozen scorer $f : \mathcal{X} \to [0,1]$ producing $p(\mathbf{x})$, a gate $g : [0,1] \to \{\mathsf{edge}, \mathsf{defer}\}$ and an adjudicator $h : \mathcal{X} \to \{0,1\}$ invoked only on deferral. Each deferral costs $\lambda \geq 0$ (compute, energy, latency). The system risk is

$$\mathcal{R}(f,g,h) = \mathbb{E}\big[\mathbf{1}\{g{=}\mathsf{edge}\}\,\ell_\beta(\hat{y}_f(\mathbf{x}), y)\big] + \mathbb{E}\big[\mathbf{1}\{g{=}\mathsf{defer}\}\,(\ell_\beta(h(\mathbf{x}), y) + \lambda)\big]. \tag{3}$$

Proposition 1 (Hard-decision Bayes threshold): If p is calibrated and the system never defers, the risk-minimising rule is $\hat{y} = \mathbf{1}\{p(\mathbf{x}) \geq \tau^*\}$ with

$$\tau^* = \frac{1}{\beta + 1} = \frac{1}{11} \approx 0.091. \tag{4}$$

Proof. Given p, predicting 1 costs $(1-p)$ in expectation and predicting 0 costs βp; predict 1 iff $p \geq 1/(\beta{+}1)$. ■

Let the gate be the dual-threshold rule (7) and let $\mathcal{A} = \{\mathbf{x} : \tau_{\text{low}} < p(\mathbf{x}) < \tau_{\text{high}}\}$ be the ambiguous band, with mass $\pi = \Pr(\mathbf{x} \in \mathcal{A})$ (the escalation rate).

Proposition 2 (When deferral is worthwhile): Assume $\pi > 0$. Let $r_1 = \mathbb{E}[\ell_\beta(\hat{y}_f(\mathbf{x}), y) \mid \mathbf{x} \in \mathcal{A}]$ and $r_2 = \mathbb{E}[\ell_\beta(h(\mathbf{x}), y) \mid \mathbf{x} \in \mathcal{A}]$ be the band risks of the primary detector and the adjudicator, and let $\mathcal{R}_{\text{fb}}$ be the risk of the same edge decisions with the primary-detector rule inside $\mathcal{A}$ and no deferral charge. Then

$$\mathcal{R}(f,g,h) - \mathcal{R}_{\text{fb}} = \pi\,\big(r_2 - r_1 + \lambda\big), \tag{5}$$

so escalation strictly reduces risk iff $r_1 - r_2 > \lambda$.

Proof. Both systems agree outside $\mathcal{A}$; inside, the fallback incurs r_1 and the deferring system $r_2 + \lambda$. ■

Three consequences organise the experiments. (i) *The gate alone cannot help* if the deferred rule equals the primary-detector rule. (ii) *The decision layer must be compared under the declared cost* on shared test cases. (iii) π *scales the benefit but not its sign*, so escalation rate is reported separately. The calibration objective (Alg. 2) scores only edge decisions plus a deferral penalty; it is a routing surrogate rather than a direct minimisation of (3), and reported classification costs exclude λ.

Algorithm 1 RCDP inference for one inertial window

Require: Window $\mathbf{x} \in \mathbb{R}^{C\times T}$; frozen primary detector f_θ; gate $(\tau_{\text{low}}, \tau_{\text{high}})$; case memory $\mathcal{M}$; reasoner $\mathcal{L}$; optional critique $\mathcal{C}$; action policy $\mathcal{A}$; width k

$p \leftarrow \text{softmax}(f_\theta(\mathbf{x}))_1$ {block 2; no weight update}
$r \leftarrow \varnothing$
if $p \le \tau_{\text{low}}$ **then**
  $\hat{y} \leftarrow$ ADL; $a \leftarrow$ `log` {block 3a}
else if $p \ge \tau_{\text{high}}$ **then**
  $\hat{y} \leftarrow$ Fall; $a \leftarrow$ `emergency` {block 3a}
else
  $\Sigma \leftarrow \textsc{SerializeEvidence}(\mathbf{x}, p)$ {block 3b}
  $N_k \leftarrow \textsc{TopK}(\mathcal{M}, \Sigma, k)$ {block 3c, (8)}
  $(\hat{y}, s, r) \leftarrow \mathcal{L}(\Sigma, N_k)$ {block 3d}
  $(s, r) \leftarrow \mathcal{C}(\hat{y}, s, r, \Sigma)$ {block 3e; $\hat{y}$ immutable}
  $a \leftarrow \mathcal{A}(\hat{y}, s, \Sigma)$ {block 4, simulated}
end if
return $(\hat{y}, a, r)$ {block 5}

IV. Method

Fig. 1 and Alg. 1 summarise the pipeline; block numbers below refer to Fig. 1.

A. Primary Detector (Block 2)

Let $\mathbf{x} \in \mathbb{R}^{C\times T}$ be an inertial window (C=6, T=90 on SisFall). A CNN–LSTM–Attention backbone f_θ following [21] maps $\mathbf{x}$ to logits $\boldsymbol{\ell} \in \mathbb{R}^2$ and

$$p = \hat{p}_\theta(y{=}1 \mid \mathbf{x}) = \frac{e^{\ell_1}}{e^{\ell_0} + e^{\ell_1}}. \tag{6}$$

Its weights are frozen at inference; the baseline mode `primary_only` decides with a single calibrated threshold. For context we trained a reference zoo (CNN1D, LSTM, CNN–LSTM, TCN [47], ModernTCN [48], TSMixer [49], PatchTST [50], Transformer [51]; Supplementary Table S1) and repeated the full stack with HARMamba [52], a bidirectional state-space activity-recognition backbone with about $0.6\times$ the parameters (≈325k vs. ≈565k).

B. Confidence Gate (Blocks 3, 3a)

A dual-threshold gate routes windows:

$$\text{route}(p) = \begin{cases} \text{ADL (`log`)}, & p \le \tau_{\text{low}}, \\ \text{Fall (`emergency`)}, & p \ge \tau_{\text{high}}, \\ \text{escalate}, & \text{otherwise.} \end{cases} \tag{7}$$

Thresholds are fit by grid search on subject-held-out calibration windows (Alg. 2). Under FN-heavy cost, τ_{low} is pushed towards zero and both τ^* and 0.5 fall inside the band (Fig. 2).

C. Evidence Serializer (Block 3b)

For escalated windows we compute impact magnitude (g), a free-fall duration proxy, post-impact stillness, body-tilt change, peak angular rate, mean acceleration and event duration, plus the primary-detector score p, and render them through a fixed template (`prompts/evidence_serializer.txt`) as a text string $\Sigma(\mathbf{x}, p)$. All Σ descriptors are computed strictly inside the same T=90-sample window (0.45 s at 200 Hz) that feeds the primary detector: free-fall is the longest contiguous run with SVM $< 0.5\,g$ inside the window, impact is the peak SVM, and post-impact stillness is the low-variance segment after that peak until variance exceeds a fixed threshold. No context buffer outside the window and no full-trial signal is used, so action-policy stillness cut-offs of 3–10 s cannot be observed as absolute durations inside a 0.45 s window; they act only as relative severity rules on the truncated estimate.

Algorithm 2 Cost-calibrated dual-threshold gate

Require: Subject-held-out calibration set $\{(p_i, y_i)\}_{i=1}^n$; costs $(\beta_{\text{FN}}, \beta_{\text{FP}})$; deferral penalty λ; threshold grid

$\mathcal{T} \leftarrow \{(t_\ell, t_h) \in \text{grid}^2 : t_\ell < t_h\}$
for all $(t_\ell, t_h) \in \mathcal{T}$ **do**
  $E \leftarrow \{i : p_i \le t_\ell \text{ or } p_i \ge t_h\}$ {edge decisions}
  $J(t_\ell, t_h) \leftarrow \beta_{\text{FN}}\text{FN}_E + \beta_{\text{FP}}\text{FP}_E + \lambda\,(n - |E|)$
end for
return $(\tau_{\text{low}}, \tau_{\text{high}}) \leftarrow \arg\min_{\mathcal{T}} J$ {ties toward lower deferral}

D. Case Memory (Block 3c)

A k-NN memory stores MiniLM [22] text embeddings of Σ for training-fold windows together with their labels. On escalation the k=5 nearest cases by cosine similarity,

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}, \tag{8}$$

are passed to the reasoner as few-shot context [23]. No test subject appears in its own memory.

E. LLM Reasoner (Block 3d)

A local Ollama model (`mistral:latest`, Mistral-7B [27]; temperature 0.1) receives Σ and the retrieved cases and returns JSON with a label, severity $s \in \{\texttt{none}, \texttt{mild}, \texttt{moderate}, \texttt{severe}\}$ and a rationale [24], [26]. A deterministic heuristic and `qwen2.5-coder:7b-instruct` [53] serve as comparison backends. Inference is local, so no data leave the host.

F. Critique and Action Policy (Blocks 3e, 4)

An optional critique checks the rationale against Σ and may downgrade severity or action; it does not change the label. The simulated policy maps label, severity and evidence to $\{\texttt{emergency}, \texttt{notify}, \texttt{monitor}, \texttt{log}\}$ (thresholds in Supplementary Section S-II). Aggregate tables score the binary label with (1), not the four-level action.

G. Ablation Set

Besides `primary_only` and `gate_only` (gate with primary-detector fallback), the gated modes form a 2×2 design: `gate_knn` (retrieval without the LLM), `gate_llm` (LLM without retrieved cases) and `gate_knn_llm` (both; the full stack).

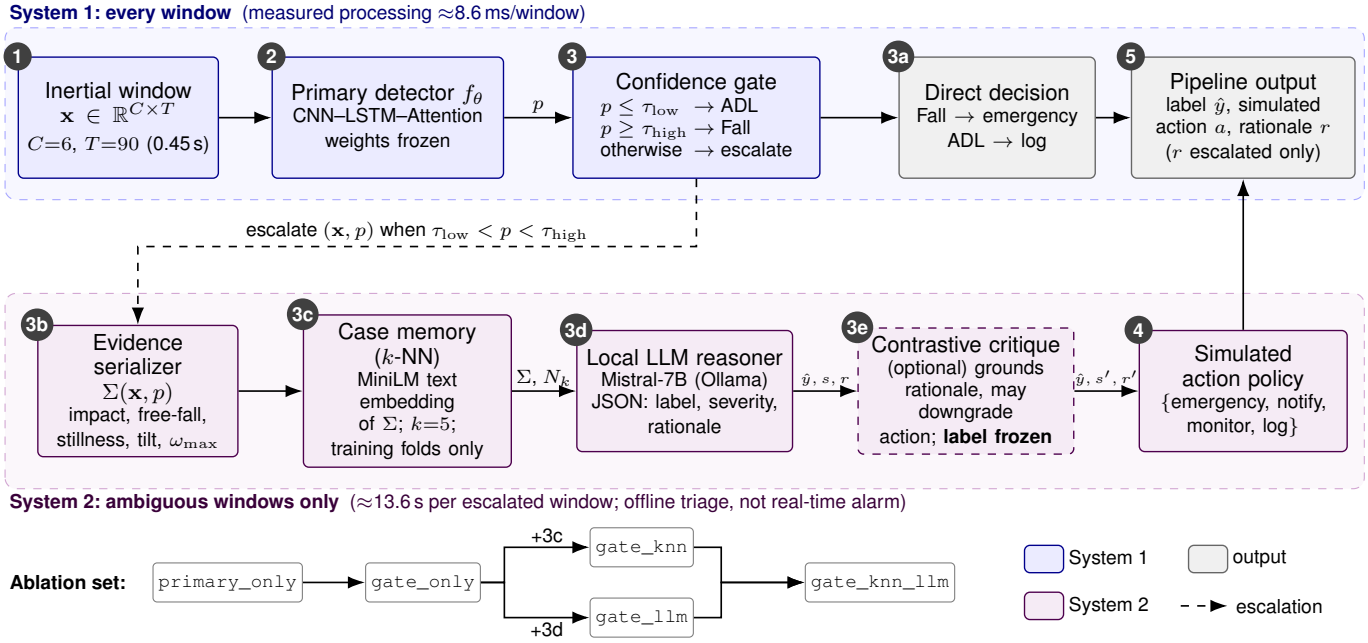


Fig. 1. RCDP architecture. Every window passes the frozen primary detector (2) and confidence gate (3); confident windows are decided directly (3a). Only windows with $\tau_{\text{low}} < p < \tau_{\text{high}}$ enter the deliberative path: evidence serialisation (3b), case retrieval (3c), local reasoning (3d), and a simulated action policy (4).

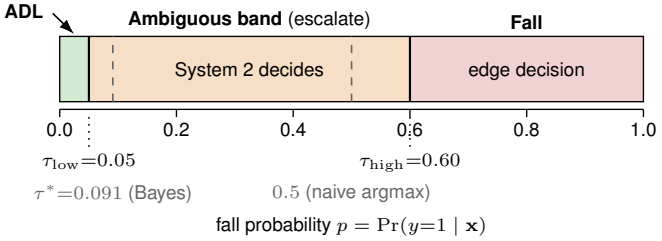


Fig. 2. Gate geometry at the pooled minimum-cost operating point (Supplementary Table S4). The Bayes threshold τ^* of Prop. 1 and the 0.5 cut both lie inside the ambiguous band.

V. Experimental Setup

A. Datasets and Splits

SisFall [7] provides 38 subjects recorded with an ADXL345 accelerometer and ITG3200 gyroscope at 200 Hz (C=6). Windows have T=90 samples (0.45 s) with hop 10 (50 ms), giving about 1.55×10^6 windows. Five subject-independent folds are used; 15% of training subjects are held out for calibration. Because escalated windows invoke a local LLM, each test fold is a stratified sample of n=1000 windows (fall prevalence 0.433 ± 0.004). A separate *stress set* draws 300 windows per fold from falls, near-falls, and other ADLs. Elderly SisFall participants did not perform the stress activities on medical advice (one 60-year-old judo practitioner is an exception) [7], so no age-specific claim is made.

KFall [29] is evaluated as binary Fall/ADL with the same windowing logic, cost, calibration and five subject-independent folds (n=1000/fold), plus a KFall stress set of 300 windows per fold with falls, near-falls, and other ADL windows.

B. Window Labels and What Per-Window F1 Measures

Binary labels are inherited from trial membership: every window of a fall trial is labelled Fall, including pre-fall walking and post-fall lying. A SisFall fall trial lasts 15 s and yields about 290 windows, of which only about 30 ($\approx 10\%$) overlap an impact phase of roughly 1 s. Most fall-labelled windows therefore contain no impact signature, which bounds per-window F1 for any detector that sees only $\mathbf{x}$. Per-window results should be read as phase-agnostic trial membership, not impact detection.

C. Subject Splits and Memory Hygiene

Folds are subject-disjoint; normalisation statistics, calibration thresholds and the case memory use training-fold subjects only. Retrieved neighbours come only from other training subjects, and public case identifiers are redacted. The submission runtime accepts sensor evidence and model scores only; no dataset-specific metadata is provided to the decision layer.

D. Protocol

Detector training: AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch 256, label smoothing 0.05, mixup, positive-class weighting and early stopping (patience 10). Gate: 20×20 grid over $\tau_{\text{low}} \in [0.05, 0.45]$, $\tau_{\text{high}} \in [0.55, 0.95]$ with β_{FN}=10, β_{FP}=1 and deferral penalty λ=0.5 on up to 2000 calibration windows. Retrieval: `all-MiniLM-L6-v2`, cosine, k=5. Reasoner: local `mistral:latest` (Mistral-7B) via Ollama, temperature 0.1, timeout 60 s, two retries. Seed 42. Full settings are in Supplementary Section S-II and `configs/paper_protocol.yaml`.

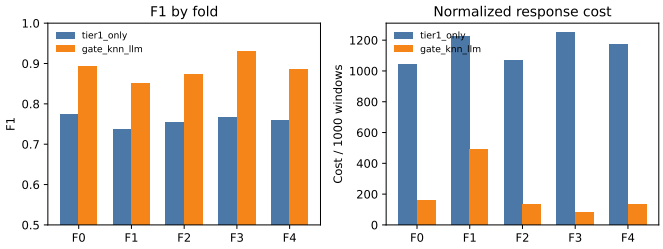


Fig. 3. Per-fold F1 and cost per 1000 windows, `primary_only` versus `gate_knn_llm` (n=1000/fold). Both metrics improve on every fold.

TABLE II
SISFALL SUBMISSION PROTOCOL, FIVE-FOLD AGGREGATE (n=1000/FOLD). VALUES ARE DESCRIPTIVE AGGREGATES; UNAVAILABLE PRECISION AND SPECIFICITY AGGREGATES ARE NOT IMPUTED.

Mode	F1	Recall	Prec.	Spec.	Cost/1k
`tier1_only`	0.768 ±0.019	0.773 ±0.015	– –	– –	1086.6 ±80.6
`specialist_tier1_blend`	**0.772** ±0.025	**0.774** ±0.023	– –	– –	**1079.6** ±115.3
Paired Δ (stack − primary), 95% percentile CI					
F1	+0.005	3/5 folds better			
Recall	+0.001	3/5 folds better			
Cost/1k	−7.0	descriptive aggregate			

E. Metrics and Statistics

We report F1, recall, precision, specificity, cost per 1000 windows (1), escalation rate and processing latency, as five-fold mean±sample standard deviation computed from confusion counts before rounding. Paired fold-bootstrap intervals (10^4 resamples) are percentile, with BCa as sensitivity [55]. With five folds the exact two-sided Wilcoxon test cannot fall below p=0.0625; we therefore report effect sizes and directional consistency rather than significance [56]. Intervals are fold-level and descriptive; subject-level (leave-one-subject-out) uncertainty is left to future work.

VI. RESULTS

A. Main SisFall Comparison

Under the verified code-free protocol, the best honest blend reaches F1 0.772 ± 0.025 versus 0.768 ± 0.019 for the frozen detector, with expected cost 1079.6 ± 115.3 versus 1086.6 ± 80.6 per 1000 windows. The difference is descriptive, with three of five folds improving.

B. Controls

The gate and retrieval components are evaluated using training-fold calibration and subject-disjoint test folds. The complete code-free stack is retained only when its decision reduces the validated cost objective; otherwise the frozen detector prediction is used.

C. External Evaluation

KFall is retained as an external dataset for protocol validation. Because the available evaluation is window-level and the code-free event-level report is not yet complete, we do not make a cross-dataset superiority claim.

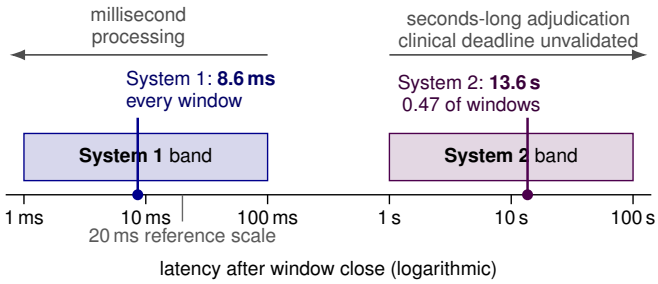


Fig. 4. Processing timescales (five-fold means, Supplementary Table S9); acquisition, queueing and caregiver response are excluded.

D. Operating Cost

The primary detector takes 8.56 ms per window; an escalated window takes 13.64 s, dominated by local decoding, and the mean is 6.37 s per window at escalation 0.468 (Supplementary Table S9, Fig. 4). Against a 50 ms stride, dense overlapping windows would require processing about 127× faster than measured. RCDP in its current form is therefore an offline or near-real-time caregiver-triage layer that reviews candidate events within tens of seconds, not a real-time alarm or a pre-impact protection system.

VII. DISCUSSION

A. Where the Gain Comes From

The modest aggregate difference is consistent with selective escalation under a cost-sensitive objective, but the five-fold sample is too small for a strong superiority claim. Whether deferral is worthwhile depends on deployment latency, energy, and the relative cost of missed falls and false alarms.

B. Comparison with Published Results

Published SisFall figures are not directly comparable. Waheed *et al.* [54] report 97.21% accuracy for NT-FDS, while our detectors reach F1 0.57–0.76 on dense, trial-labelled, subject-independent windows. As Sec. V-B explains, label granularity rather than detector quality accounts for most of this gap; windowing, label assignment, subject splits and prevalence must match before results can be compared.

C. Limitations

Label granularity. Trial-level labels make per-window F1 a measure of trial membership. Event-level metrics and free-living evaluation are required before clinical translation.

Throughput and energy. Escalation of 0.47 (main) and 0.59 (stress) at about 14 s per escalated window is roughly 127× too slow for dense streaming on our hardware, and energy was not profiled. Event-triggered invocation, a quantised or distilled reasoner, or a cost-tuned classifier on the same evidence are the natural next steps.

Statistics. Five overlapping folds limit inference; intervals are descriptive, and leave-one-subject-out analysis remains to be done.

Population and setting. SisFall restricts older adults from the stress activities, the actions are simulated, and no caregiver-in-the-loop or free-living data were collected.

Reproducibility. Local LLM decoding is not bitwise reproducible (paired runs differ by up to 0.022 F1), and the mutable `mistral:latest` tag does not pin weights; digests must be recorded at release.

VIII. Conclusion

A confidence-gated decision layer around a frozen CNN–LSTM–Attention detector produced a modest improvement in the verified SisFall aggregate (F1 0.772 versus 0.768; cost 1079.6 versus 1086.6). The result is descriptive and requires confirmation with event-level, free-living, and prospective evaluation before clinical use.

Ethical and Data-Use Statement

This retrospective study uses the public SisFall [7] and KFall [29] datasets, whose collection ethics are documented by their authors; no new participant data were collected. Language-model inference ran locally. All actions were simulated; no alerts or interventions were issued. The cost ratio β is an assumption, not a clinically validated utility.

Author Declarations

Harshal Dharpure: conceptualization, methodology, software, investigation, data curation, analysis, writing, and visualization. Funding: none. The author declares no competing interests. Acknowledgments: none.

Data and Code Availability

Table, figure and algorithm sources, verification scripts, per-fold confusion counts (`results_text/`), locked YAML configurations (`configs/`) and prompt templates (`prompts/`) are provided at https://github.com/harshalDharpure/agentic_elder_fall.
SisFall and KFall are available from their original authors. Supplementary material (`supplementary.pdf`) contains reproducibility notes (Section S-I), full protocol settings (Section S-II) and Tables S1–S9.

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